

Zhiyu Wu

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Education

University of Illinois Urbana-Champaign

Ph.D. Computer Science

Advisor: Prof. Fan Lai

Research Interest: Systems for ML, Inference/Training/RL software stacks, Efficient ML

Champaign, IL

Aug 2026 – present

University of Illinois Urbana-Champaign

M.S. Computer Science – Thesis Based

Champaign, IL

Aug 2024 – Dec 2025

University of Michigan

B.S.E. Computer Engineering (Dual Degree)

Ann Arbor, MI

Aug 2022 – May 2024

Shanghai Jiao Tong University

B.S.E. Electrical and Computer Engineering (Dual Degree)

Shanghai, China

Sept 2020 – Aug 2024

Publications

- [JITServe: SLO-aware LLM Serving with Imprecise Request Information](#); NSDI, 2026; **Zhiyu Wu***, Wei Zhang*, Yi Mu, Rui Ning, Banruo Liu, Nikhil Sarada, Myungjin Lee, Fan Lai
- [CacheFlow: Efficient LLM Serving with 3D-Parallel KV Cache Restoration](#); Preprint, 2026; Sean Nian, Hanzhang Shen, **Zhiyu Wu**, Jiahao Fang, Qilong Feng, Fan Lai
- [Andes: Defining and Enhancing Quality-of-Experience in LLM-Based Text Streaming Services](#); Preprint, 2024; Jiachen Liu, **Zhiyu Wu**, Jae-Won Chung, Fan Lai, Myungjin Lee, Mosharaf Chowdhury
- [The ML. ENERGY Benchmark: Toward Automated Inference Energy Measurement and Optimization](#); NeurIPS, spotlight, 2025; Jae-Won Chung, Jiachen Liu, Jeff J Ma, Ruofan Wu, Oh Jun Kwon, Yuxuan Xia, **Zhiyu Wu**, Mosharaf Chowdhury

Work Experience

Software Engineering, Bytedance, AML

San Jose, CA

Supervisor: Xiao Yu, Zilin Xu

Jan 2026 – Aug 2026

- Optimize training performance of [VeOmni](#) on B200, including framework and kernel optimizations.
- Work on Post-training pipeline and data plane of LLM training.
- Establish TPU inference stack with colleagues and collaborators from Google, especially, working on KV cache management, DP/TP, MTP adaption, and adaption to Seed models.

Research Engineering Intern, Bytedance, AML

San Jose, CA

Supervisor: Xiao Yu

May 2025 – Aug 2025

- Built end-to-end AI model training workflow with Jupyter notebook and web frontend, encompassing dataset profiling, training parameter optimization, SFT/RL deployment, evaluation, and production inference pipeline.
- Contributed to LLM training simulator with memory analysis, communication modeling, and MFU metrics.
- Explore and optimize memory usage for SFT/pretrain with FSDP, specifically, optimize compiler/kernel to maximum tensor reuse in backward and optimizer stage.

Research Experience

CacheFlow: Efficient LLM Serving via Automated 3D-Parallel KV Cache Restoration

Champaign, IL

Supervisor: Fan Lai

Aug 2024 – Sept 2026

- Formulated KV cache restoration as a multidimensional scheduling problem, coordinating recomputation and cache loading across tokens, layers, and GPUs.
- Developed a batch-aware dynamic-programming planner that jointly selects recomputation boundaries across requests and layers, accounting for heterogeneous full-attention and recurrent-layer costs.
- Enabled concurrent restoration across model shards by caching lightweight boundary activations, removing sequential recomputation dependencies between GPUs.

- Implemented CacheFlow atop vLLM and LMCache, reducing TTFT by 10%–62% across diverse models, chatbot and agentic workloads, and hardware configurations.

JITServe: SLO-aware LLM Serving with Imprecise Request Information

Champaign, IL

Supervisor: Fan Lai

Jul 2024 – Sept 2025

- Developed an SLO-aware scheduler for latency-sensitive, throughput-sensitive, and compound requests, allocating resources to meet heterogeneous performance requirements.
- Represented multi-agent workflows as DAGs and applied graph matching to predict execution dependencies and remaining workload; used quantile regression forests to estimate response lengths.
- Designed service-density-based prioritization to maximize service gain while preserving capacity for other requests and preventing starvation of lower-priority workloads.
- Achieved 1.3–8.3× higher service gain and 4.0–10.3× higher SLO goodput than existing serving systems across diverse workloads and models.

Andes: Defining and Enhancing Quality-of-Experience in LLM-Based Text Streaming Services

Ann Arbor, MI

Supervisor: Mosharaf Chowdhury, Amber Liu

May 2023 – April 2024

- Identified the mismatch between conventional serving metrics and streaming user experience, highlighting the importance of delivering text at a pace that matches user reading speed.
- Defined a Quality-of-Experience (QoE) metric that tracks token delivery throughout generation and captures its impact on the user’s reading experience.
- Formulated online scheduling as a knapsack optimization problem, allocating serving resources to maximize overall QoE across concurrent requests.
- Built Andes atop vLLM and integrated the QoE-aware scheduler into the serving runtime to improve the responsiveness of LLM text streaming.

Project Experience

Symbiotic Lab/ML.ENERGY.LEADERBOARD Team

Ann Arbor, MI

Developer

May 2023 – Sept 2023

- Developed the ML.ENERGY Leaderboard, an open-source platform for benchmarking the energy efficiency and NLP performance of LLM models.
- Defined performance metrics and implemented scripts for optimized batched inference to ensure accurate measurement.
- Contributed to the online Chatbot Arena which gathers data on models' energy consumption and performance.

Embedded Device for Keystroke Timing and Acoustic Attack Protection

Ann Arbor, MI

- Designed the device to intercept keystrokes and introduce random delays before sending to the PC.
- Implemented keystroke sound playback to counter acoustic attacks using recorded sounds.
- Based the system on the STM32F405 microcontroller with Embedded Rust for secure and efficient performance.
- Delivered a compact, user-friendly design with production costs around \$24.50 per unit.
- Utilized SD card for storing custom keystroke sounds and USB peripherals for communication with keyboard and host PC.

Professional Service

- VP 160 Honors Physics, SJTU
- Volunteer teacher for Mine Village Primary School, Luxi Town, Yunnan Province, China.

Summer 2022

Dec 2021 – Jan 2022

Honors

Dean List, <i>Umich</i>	2023
University Honor, <i>Umich</i>	2022
Tang Junyuan Scholarship, <i>SJTU</i>	2022
SJTU Undergraduate Excellent Scholarship Class B, <i>SJTU</i>	2022